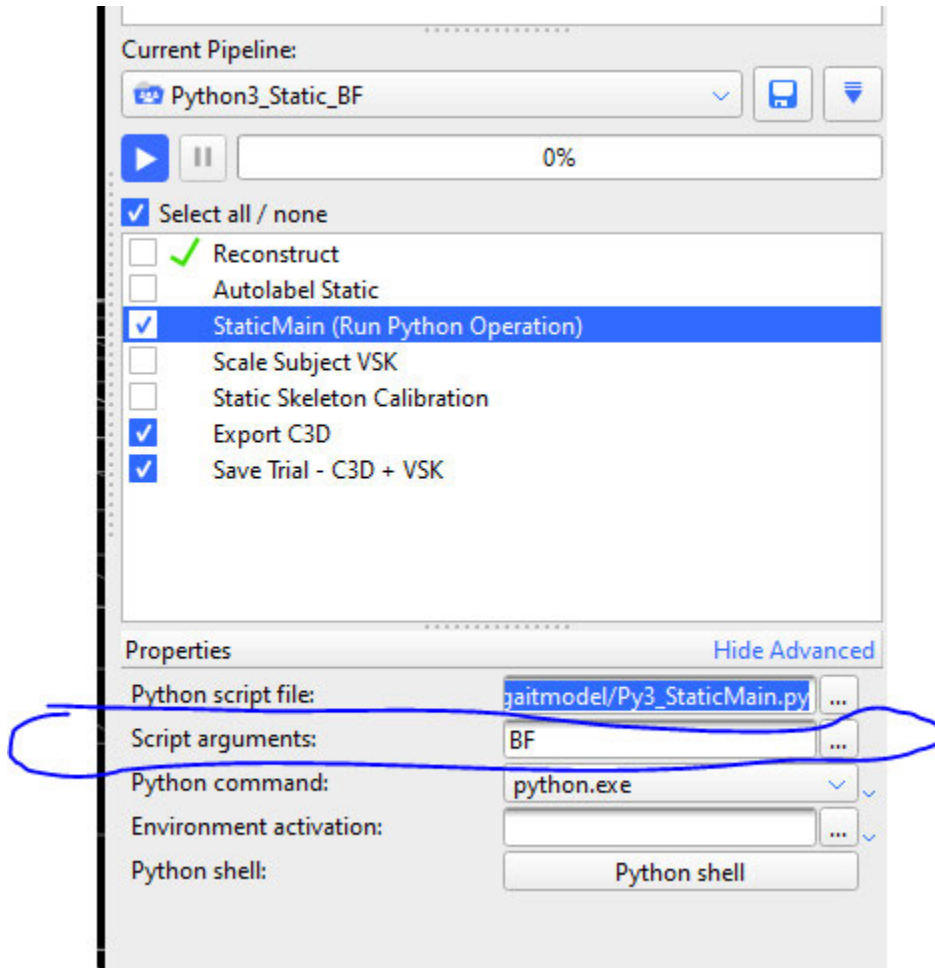


Use of Script Arguments in Py3_StaticMain.py, Py3_DynamicMain.py and Py3_CreateGCD.py

1. In the typical workflow process, the StaticMain (Py3_StaticMain.py) maintains a subscript to link static trial information to be used in dynamic walking trial calculation.
 - a. Typically, the first condition uses the subscript BF (for BareFoot) to link all BF data collection.
 - b. The image below indicates the location of the Script argument.



2. After executing the StaticMain pipeline command the following screen appears. The typical entries for Barefoot Data collection are entered. The screen shot below included a foot model.

Static

First Name

Last Name

Patient Number

Date Of Birth

01

Jan

1981

Trial Activity

Static

TrialModifier

Barefoot

Static Forward Direction

-X

AssistiveDevice

None

Static Frame Number

21

Static File

Anthropometric File

-

Date Of Data Capture

09

Jan

2024

kg

74.3

BodyMass

mm

1870

Height

mm

191

ASIS-to-ASIS Distance

mm

Left

920

Right

915

Leg Length

mm

109

108

Knee Width

mm

81

83

Ankle Width

mm

69

69

ASIS-to-GT Distance

Foot Model

☒ Left

☒ Right

deg

0

4

Hindfoot Varus/Valgus

deg

Calcaneal Pitch

deg

First Metatarsal Pitch

Special Test Conditions

☒ Left

☒ Right

Subject Plantigrade during Static Trial

☐ Subject Shod (with or without orthoses)

mm

Left

Right

Difference in sole thickness between heel and toes (mm)

Knee Option

☐ KAD

☒ M/L Markers

☒ N/A

☐ Iliac

☐ Triad

Apply Pelvic Fix Option

Save & Proceed

Quit

3. The standing posture angles for the typical Barefoot condition appear after 2 screens.

Static

Static File-
Model-
User Preference

Standing Posture (degrees)

	Left	Right
Trunk Obliquity	2 Down	2 Up
Trunk Tilt	6 Back	6 Back
Trunk Rotation	6 Ext	6 Int
Pelvic Obliquity	0	0
Pelvic Tilt	9 Ant	9 Ant
Pelvic Rotation	4 Ext	4 Int
Hip Ab/Adduction	3 Abd	2 Abd
Hip Flex/Extension	8 Flex	7 Flex
Hip Int/Ext Rotation	4 Ext	9 Ext
Knee Varus/Valgus	1 Var	1 Var
Knee Flex/Extension	5 Flex	5 Flex
Knee Int/Ext Rotation	17 Ext	18 Ext
Knee Progression	8 Ext	5 Ext
Ankle Plantar/Dorsiflexion	7 Dorsi	7 Dorsi
Ankle Int/Ext Rotation	14 Int	11 Int
Foot Progression	12 Ext	13 Ext

Foot Segments

Hindfoot Pitch	11 Up	16 Up
Hindfoot Progression	14 Ext	6 Ext
Hindfoot Varus/Valgus	0	4 Val
Forefoot Pitch	25 Down	23 Down
Forefoot Progression	4 Ext	7 Ext
Midfoot Complex Ab/Adduction	10 Add	1 Abd
MTP1 Varus/Valgus	11 Val	8 Val

Tibial Torsion (degrees)

16 Ext	18 Ext
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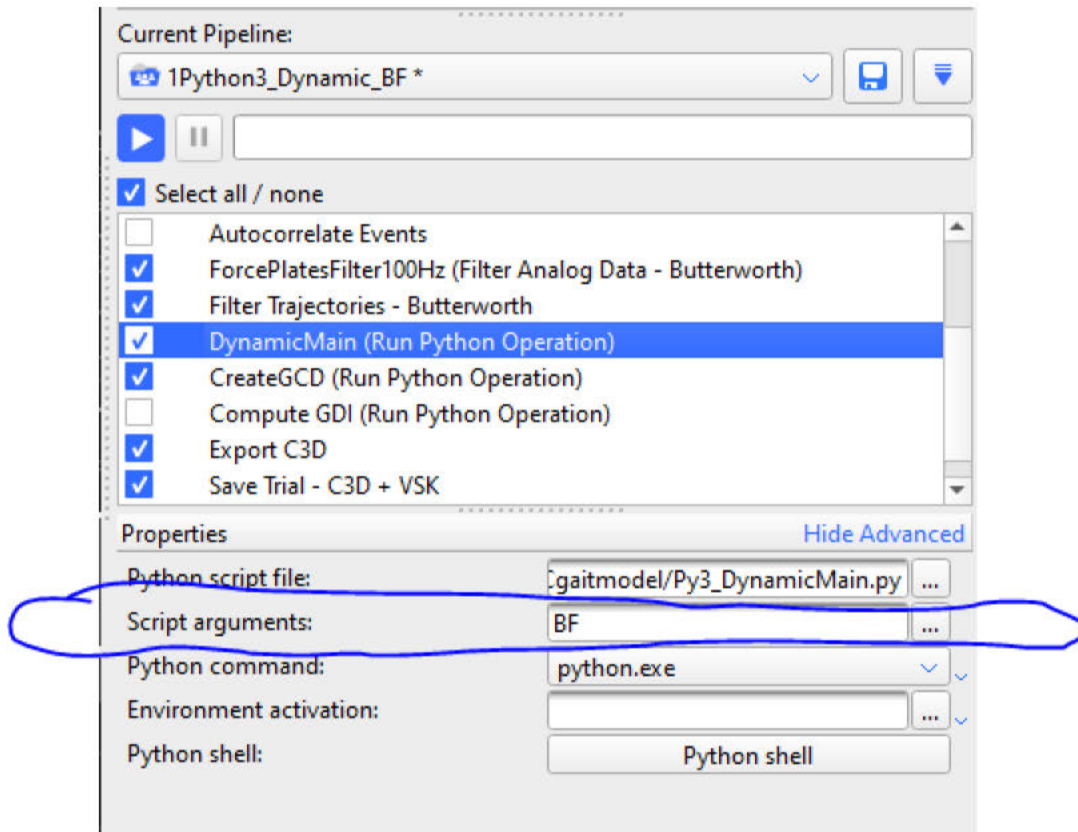
Save PDF

Save Results

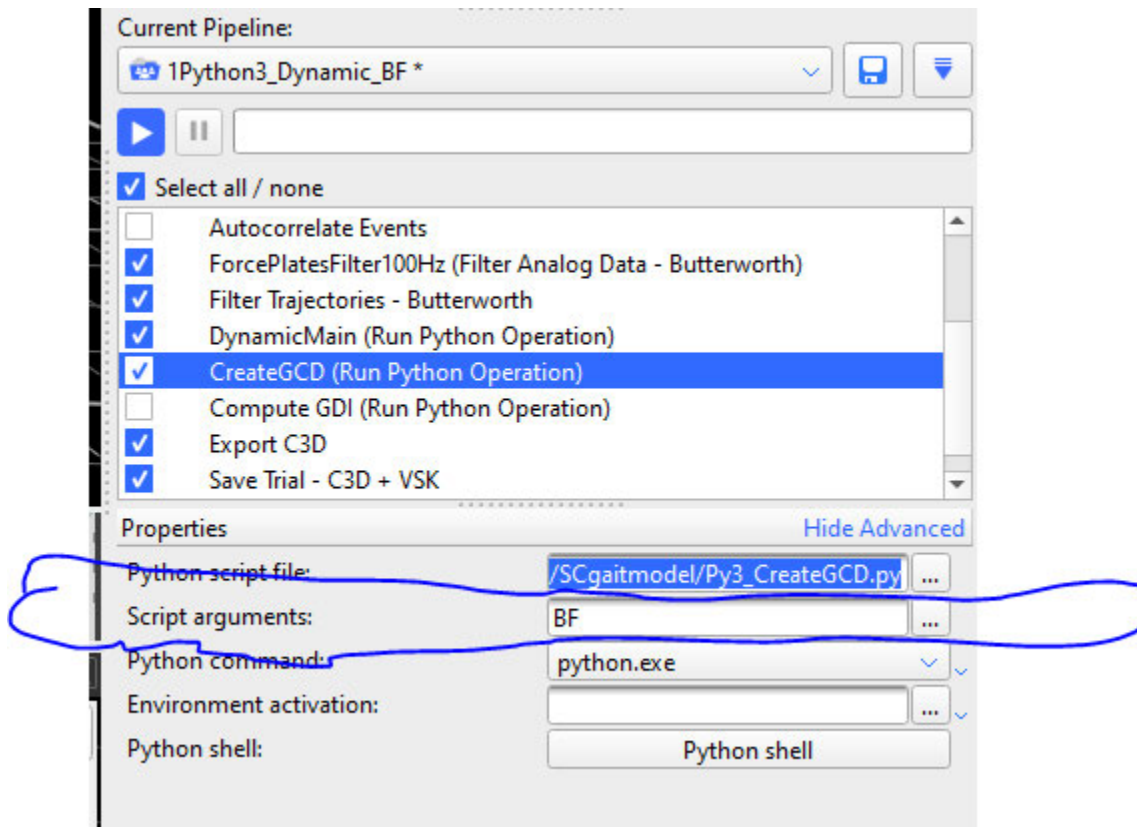
Back

Quit

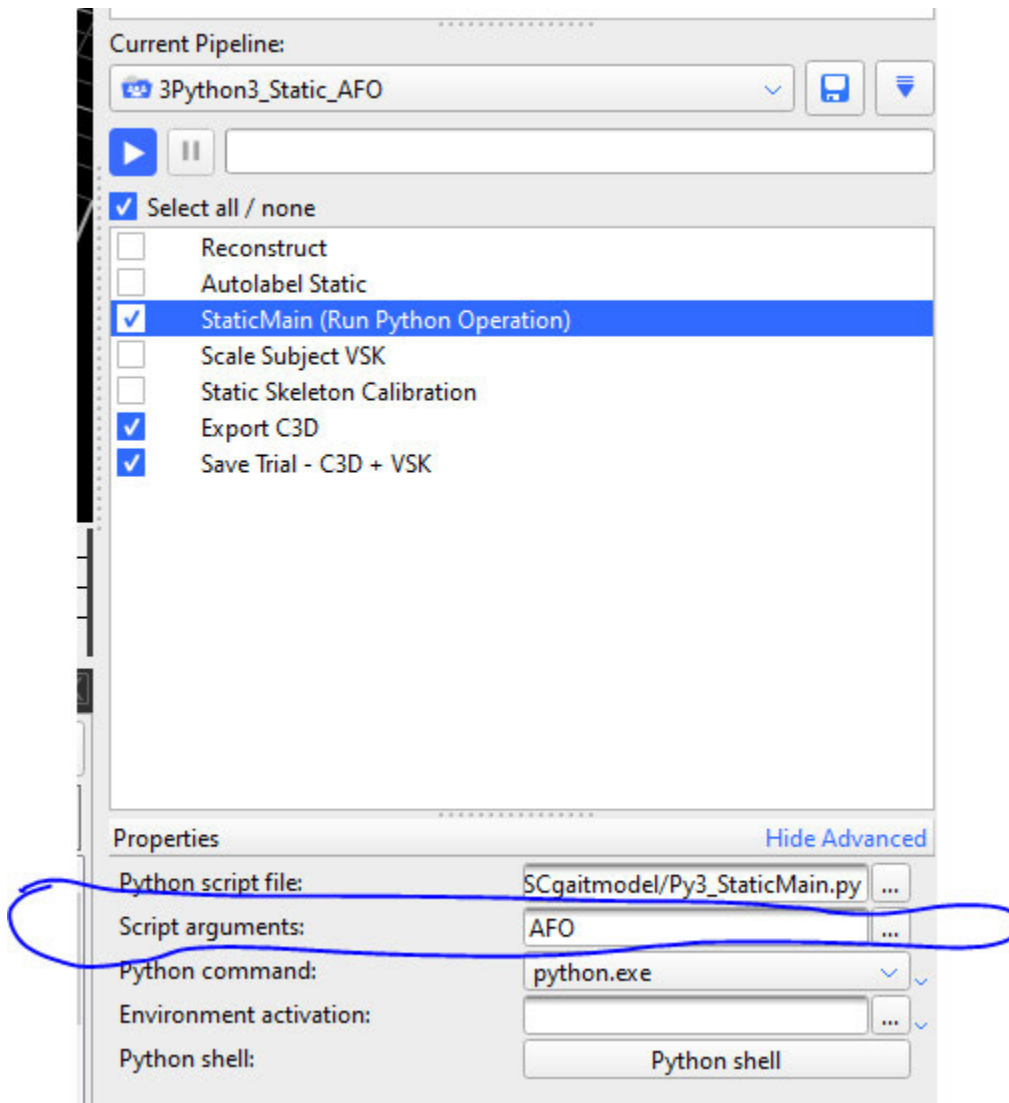
4. After selecting the “Save Results” and “Quit” buttons, the transformation matrices are saved in the follow file: “Static_BF_██████████.py”
- Note the file name contains the subscript “BF” so that future processing commands with the “BF” subscript will refer to this file.
 - When the DynamicMain pipeline command and CreateGCD pipeline commands are executed (Py3_DynamicMain.py and Py3_CreateGCD.py), the script argument needs to match the Static Script argument.
 - NOTE: The Script argument in the DynamicMain command needs to match the appropriate Static Script argument, usually BF in the initial steps.



- d. NOTE: Likewise, the Script argument in the CreateGCD command needs to match the appropriate Static Script argument, usually BF in the initial steps.



5. When another condition is tested that requires a new static trial is started, the Script arguments should be modified to fit the new condition. For example, if data is collected for a patient wearing braces, a new static trial is needed (the markers configuration is changed and ankle width is a different value). However, if a condition such as the use of assistive device (or no assistive device) is tested, there is no need for a new static trial so there is not a revision of the Script arguments.
- Suppose a Brace condition is tested. The StaticMain (Py3_StaticMain.py) command should be revised to a new Script argument. In this case a subscript such as "AFO" would be appropriate to link static trial information to be used in dynamic walking Brace trial calculations. See Screen shot example below:



- b. When a the Py3_StaticMain.py command is executed for a Brace condition, it appropriate to revise several entries. 1) TrialModifier, 2) Ankle Width, 3) Foot Model unchecked, 4) Subject shod with Heel-Toe height differences. See screen shot below:

The screenshot shows the Static software interface with the following sections and annotations:

- Patient Information:**
 - First Name: [Empty]
 - Last Name: [Empty]
 - Patient Number: MooreJackson
 - Date Of Birth: 01 Jan 1981
 - Trial Activity: Static
 - TrialModifier: B AFO-Solid (Annotated: "Add Brace Condition here")
 - Static Forward Direction: -X
 - AssistiveDevice: None
- Anthropometric Parameters:**
 - Static File: MooreJ12
 - Anthropometric File: [Empty]
 - Date Of Data Capture: 09 Jan 2024
 - kg: 74.3
 - mm: 1870
 - mm: 191
 - Left: 920, 109, 100, 69
 - Right: 915, 108, 100, 69
 - BodyMass
 - Height
 - ASIS-to-ASIS Distance
 - Leg Length
 - Knee Width
 - Ankle Width (Annotated: "Ankle width is likely to be different with a brace")
 - ASIS-to-GT Distance
- Foot Model:**
 - Left: 0
 - Right: 4
 - Hindfoot Varus/Valgus
 - Calcaneal Pitch
 - First Metatarsal Pitch
- Special Test Conditions:**
 - Left: [checked], Right: [checked]
 - Subject Plantigrade during Static Trial
 - Subject Shod (with or without orthoses) [checked]
 - Left: 10, Right: 10
 - Difference in sole thickness between heel and toes (mm)
 - Knee Option: KAD, M/L Markers
 - N/A, Iliac, Triad
 - Apply Pelvic Fix Option

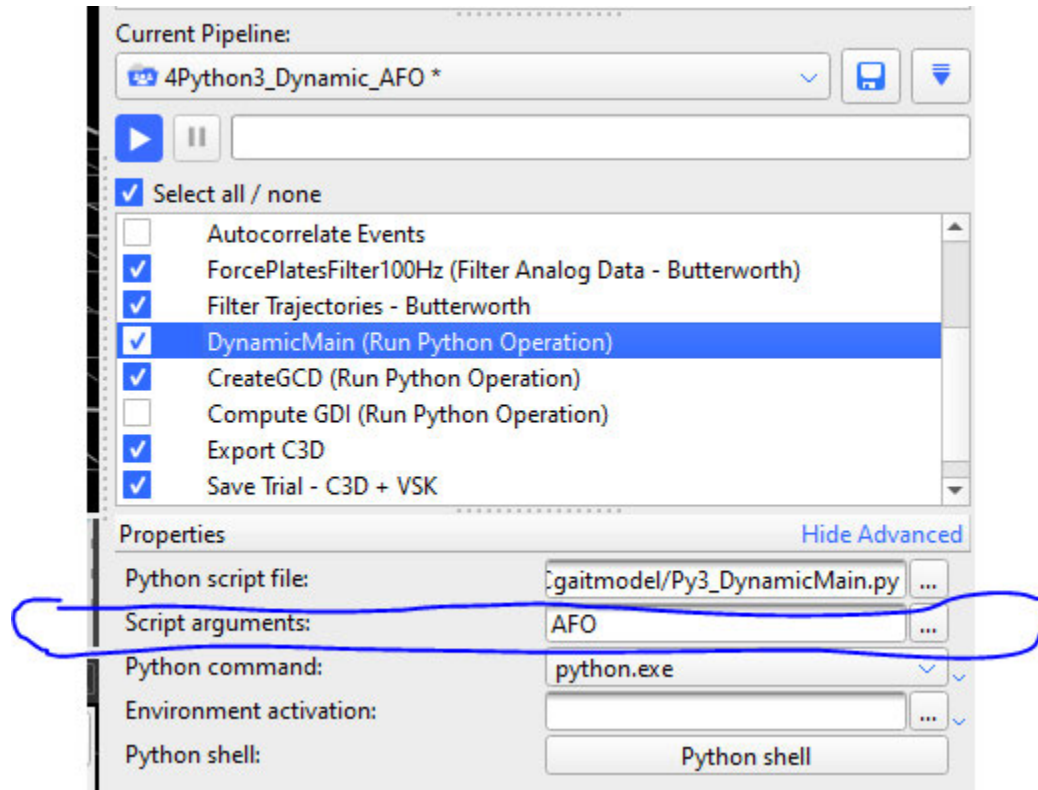
Annotations:

- "Add Brace Condition here" points to TrialModifier: B AFO-Solid.
- "Ankle width is likely to be different with a brace" points to Ankle Width.
- "This should be unchecked for Braces, even if a foot model was used in the BF condition, a change in VST will uncheck this" points to the Foot Model section.
- "Check, subject shod, enter the measure difference in heel and toe height here for Shoes/Braces." points to the Subject Shod and sole thickness difference fields.

- c. The typical 2 screens following in the StaticMain.py execution. After selecting the "Save Results" and "Quit" buttons, the transformation matrices are saved in the follow file:

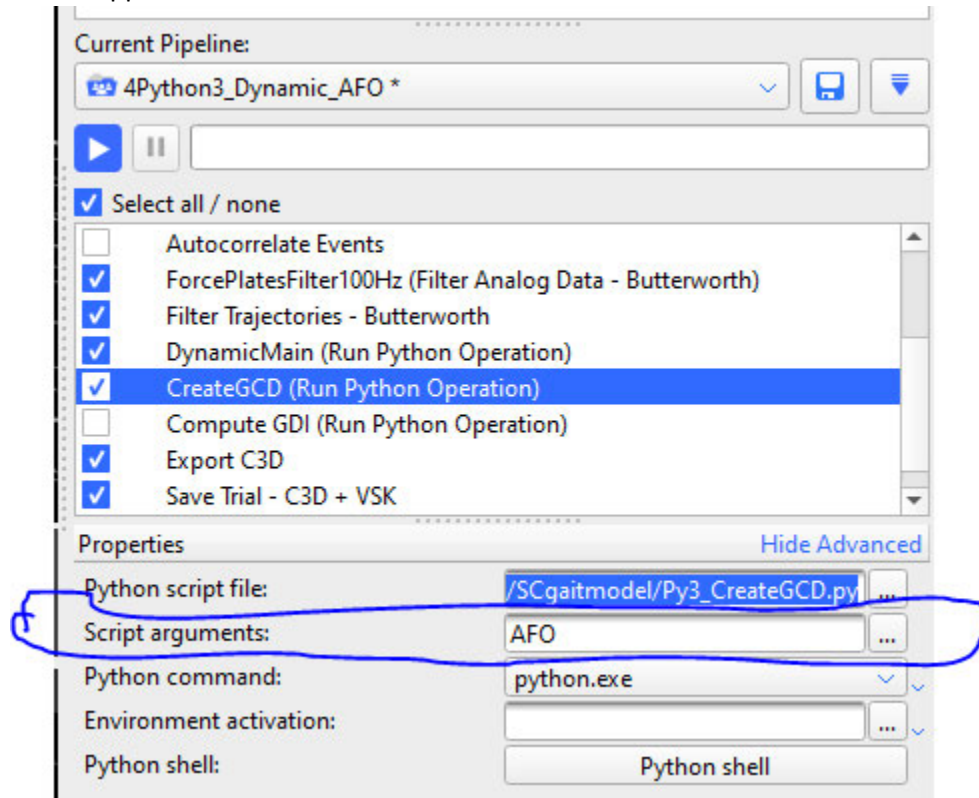
"Static_AFO-[Redacted].py"

- d. Note the file name contains the subscript “AFO” so that future processing commands with the “AFO” subscript will refer to this file.
- e. When the DynamicMain pipeline command is executed (Py3_DynamicMain.py), the following screen appears



- f. NOTE: The Script argument in the DynamicMain command needs to match the appropriate Static Script argument, AFO in this case if it is used for Brace trials

- g. When the CreateGCD pipeline command is executed (Py3_CreateGCD.py), the following screen appears:



- h. NOTE: Likewise, the Script argument in the CreateGCD command needs to match the appropriate Static Script argument, AFO in this case if it is used for Brace trials

6. Implementation Notes:

- a. The Script arguments can be generic such as condition 2 (Cond2), condition 3 (Cond3), and so on.
- b. Multiple pipelines for Python3_Static and Python3_Dynamic are appropriate for conditions to maintain consistent subscripts.
 - i. For example, pipelines titled Python3_Static_Cond2 and Python3_Dynamic_Cond2 can be created with revised Script arguments to be used for the first non-barefoot condition
 - ii. pipelines titled Python3_Static_Cond3 and Python3_Dynamic_Cond3 can be created with revised Script arguments to be used for the another condition